READING PASSAGE 1

*You should spend about 20 minutes on **Questions 1–13**, which are based on Reading Passage 1 below.*

Shipwreck in the Forbidden Zone

Five centuries ago a ship loaded with gold wrecked off a beach laden with diamonds.

History rarely unfolds like a fable. But consider this: A 16th-century Portuguese trading vessel, carrying a fortune in gold and ivory and bound for a famed spice port on the coast of India, is blown far off course by a fierce storm while trying to round the southern tip of Africa. Days later, battered and broken, the ship founders on a mysterious, fogbound coast sprinkled with more than a hundred million carats of diamonds, a cruel mockery of the sailors' dreams of riches. None of the castaways ever return home.

This improbable yarn would have been lost forever had it not been for the astonishing discovery in April 2008 of a shipwreck in the beach sands of the Sperrgebiet—the fabulously rich and famously off-limits De Beers diamond-mining lease near the mouth of the Orange River on Namibia's southern coast. A company geologist working in mining area U-60 came across what at first he took to be a perfectly round half sphere of rock. Curious, he picked it up and immediately realized it was a copper ingot. A strange trident-shaped mark on its weathered surface turned out to be the hallmark of Anton Fugger, one of Renaissance Europe's wealthiest financiers. The ingot was the type traded for spices in the Indies in the first half of the 16th century.

Archaeologists would later find a staggering 22 tons of these ingots beneath the sand, as well as cannon and swords, ivory and astrolabes, muskets and chain mail—thousands of artifacts in all. And gold, of course, fistfuls of gold: more than 2,000 beautiful, heavy coins—mainly Spanish excelentes bearing the likenesses of Ferdinand and Isabella, but also a smattering of Venetian, Moorish, French, and other coinage, as well as exquisite portugueses with the coat of arms of King João III.

It is by far the oldest shipwreck ever found on the coast of sub-Saharan Africa, and the richest. Its dollar value is anyone's guess, but none of its treasures have fired the imaginations of the world's archaeologists as much as the wreck itself: a Portuguese East Indiaman from the 1530s, the heart of the age of discovery, with its cargo of treasure and trade goods intact, having lain untouched and unsuspected in these sands for nearly 500 years.

"This is a priceless opportunity," says Francisco Alves, the doyen of Portuguese maritime archaeologists and the head of nautical archaeology under the Ministry of Culture. "We know so little about these great old ships. This is only the second one ever excavated by archaeologists. All the others were plundered by treasure hunters."

Treasure hunters are never going to be a problem here, not in the middle of one of the world's most jealously guarded diamond mines, on a coast whose very name—Sperrgebiet—means "forbidden zone" in German. Far from plundering, officials at De Beers and in the Namibian government, who work the lease as a joint venture called Namdeb, suspended their operations around the wreck site, called in a team of archaeologists, and for a few gloriously diverting weeks mined history instead of diamonds.

It will take scholars years to study the wealth of material gleaned from the Diamond Shipwreck, as it has come to be called. "So much is unknown," says Filipe Vieira de Castro, the Portuguese-born coordinator of the nautical archaeology program at Texas A&M University. Castro has spent more than ten years studying Portuguese trading vessels, or naus, lately developing computer models based on the slender archaeological pickings available. "This wreck will give us new insights into everything from hull design, rigging, and how these ships evolved, to little day-to-day things such as how they cooked meals on board and what people brought with them on these great journeys."

Already, some inspired detective work among the rare manuscripts and royal archives in Lisbon has cobbled together enough bits and pieces to tell the tale of a long-forgotten voyage and a vanished ship that turned out to be as rich in irony and allegory as it was in gold.

The story begins on a fresh spring day in Lisbon—Friday, the seventh of March, 1533, to be exact—when the great naus of that year's India fleet sailed grandly down the Tagus River and out into the broad Atlantic, flags and pennants flying and colorful silks and velvets draped from their towering castles. These were the pride of Portugal, the space shuttles of their day, off on a 15-month odyssey to bring back a fortune in pepper and spices from distant continents. Goa, Cochin, Sofala, Mombasa, Zanzibar, Ternate: Storied places that once had been as remote as the stars were now familiar ports of call, part of the Portuguese vernacular, thanks to Portuguese ingenuity and cutting-edge technology.

The outbound ships that sailed down the Tagus River in 1533 were sturdy and capable; two of them were brand-new and owned by the king himself. One of these was the *Bom Jesus—the Good Jesus*—captained by one Dom Francisco de Noronha and carrying 300 or so sailors, soldiers, merchants, priests, nobles, and slaves.

PINNING A NAME and a story to an anonymous, five-centuries-old shipwreck found unexpectedly on a far-flung shore takes canny sleuthing and more than a little luck—particularly if it is thought likely to have been an early Portuguese wreck. Although the Spanish Empire left mountains of paperwork in its wake, a catastrophic earthquake, tsunami, and fire in November 1755 virtually wiped Lisbon off the map and sent the Casa da India, the building that housed the vast majority of precious maps, charts, and shipping records, tumbling into the Tagus River.

"That left a huge hole in our history," says Alexandre Monteiro, a maritime archaeologist and researcher who works with the Portuguese Ministry of Culture. "With no India archives left to peruse, one has to revert to other, more imaginative ways of finding information."

In this instance, a vital clue came from the coins found in abundance on the wreck—particularly those beautiful and rare portugueses of King João III. These were minted for only a few years, from 1525 to 1538, after which they were recalled, melted down, and never reissued. Finding so many sparkling new portugueses on the wreck is a strong indication that the ship sailed during this 13-year window in time. Moreover, the load of copper ingots suggests the ship was on its outward passage to India to buy spices rather than returning.

Although the complete Casa da India records are long gone, some tantalizing snippets remain in libraries and archives that survived the 1755 earthquake. Among these are the *Relações das Armadas*, the so-called narratives of the fleets. A thorough study of the most complete narratives shows that 21 ships were lost on the way to India between 1525 and 1600. Only one of these went down anywhere near Namibia: the *Bom Jesus*, which sailed in 1533 and was "lost on the turn of the Cape of Good Hope."

Another intriguing pointer to the *Bom Jesus* comes from a letter Monteiro unearthed in the royal archives. Dated February 13, 1533, it reveals that King João had just sent a knight to Seville to pick up 20,000 crusadoes' worth of gold from a consortium of businessmen who had invested in the fleet that was about to sail for India—the fleet that included the *Bom Jesus*. Archaeologists had been puzzled by the huge quantity of Spanish coins found among the wreckage—about 70 percent of the gold pieces were excelentes, unexpected for a Portuguese ship. "This letter would go a long way toward explaining that," says Monteiro. "Spanish investors, it seems, had an unusually large stake in the 1533 fleet."

A rare 16th-century tome called the Memória das Armadas even offers a tantalizing glimpse of the *Bom Jesus*. Issued as a commemorative volume, a sort of Renaissance-era coffee-table book, it contains illustrations of all the fleets that sailed for India each year after Vasco da Gama pioneered the route in 1497. Among the pictures for 1533 is a vignette of two rigged masts under full sail disappearing into the waves and the words *"Bom Jesus"* together with a simple epitaph: perdido—lost.

So what did happen? It seems that four months or so after its grand departure from Lisbon, the first fleet of 1533 was struck and scattered by a huge storm. Details are sketchy. An account of the voyage by Captain Dom João Pereira, the fleet's commander, has been lost. All that remains is a clerk's acknowledgement that the report was received and a mention that the *Bom Jesus* disappeared in wild weather somewhere off the cape. It is easy to envision what might have happened next: The storm-battered ship was caught up in the powerful winds and currents that surge along the southwest African coast and was driven helplessly northward for hundreds of miles. As the windswept scrub of the Namib Desert hove into view, the doomed nau struck an outcrop of rock about 150 yards from shore. The shuddering blow broke off a big chunk of the stern, spilling tons of copper ingots into the sea and sending the *Bom Jesus* to its grave.

FAST-FORWARD five centuries to a maritime archaeology site that feels slightly surreal. A knot of researchers in hats and sunscreen are excavating a sunken ship that rests some 20 feet below sea level, the Atlantic Ocean held back by a massive earthen retaining wall that leaks a bit along its base. Closed-circuit television cameras, set up around the perimeter of the site, monitor everyone's movements—a reminder that for all the excitement of the find, this is still a diamond mine. And a rich one, where loose diamonds could well be mingled in the sands the archaeologists are brushing away.

"If it hadn't been for those copper ingots weighing everything down, there would be nothing left here to find," says Bruno Werz, director of the Southern African Institute of Maritime Archaeology, who was called in from Cape Town to assist with the excavation. "Five centuries of storms and waves would have washed everything away."

Werz and a team of researchers have been poring over the wreckage, measuring, photographing, scanning the site millimeter by millimeter with a state-of-the-art, three-dimensional laser scanner. They are trying, among other things, to piece together the ship's final harrowing moments, which would not have been pretty—the mangled remains of the hull and forecastle and a tangle of sails, spars, and rigging sloshing about in the swell, drifting north with the current and probably breaking apart as it went. Mine workers found a huge wooden rigging block three miles farther up the coast.

And what of the people on board, Dom Francisco and the rest?

"A winter storm along this coast is no joke," says Dieter Noli, the mine's resident archaeologist, who has lived and worked along this stretch of the Namib Desert for more than ten years. "It would have been nasty, with winds of over 80 miles an hour and a huge breaking surf. Getting ashore would have been just about impossible. On the other hand, if the storm had blown itself out and the ship wallowed ashore on one of those quiet, fog-shrouded days we also get around here, well, now that opens up all kinds of interesting possibilities."

That may have happened. Although the discovery of human toe bones in a shoe found pinned beneath a mass of timbers indicates that at least one person did not survive, those were the only human remains recovered from the wreck. And few personal possessions were found among the artifacts. These facts lead archaeologists to believe that despite the breakup of the ship along the surf line, many if not most of those aboard made it to land.

And then what? This is one of the most inhospitable places on Earth, an uninhabited wasteland of sand and scrub stretching for hundreds of miles. It was winter. They were cold and wet, exhausted and bereft. There was no hope of rescue or a search party, for nobody in the outside world knew they were alive, let alone where to start looking. Nor was any ship likely to pass this way by chance; they were far off the trade routes. As for somehow getting back to Portugal—well, the crew might as well have been shipwrecked on Mars.

All the same, things needn't necessarily have ended badly for the castaways, according to Noli. The Orange River lay only 16 miles to the south of the wreck, a source of fresh water whose bloom they might have noticed as they drifted by its mouth. And there was plenty of food about: shellfish, seabird eggs, and loads of desert land snails.

What's more, the Portuguese could have met the local survival experts. Winter was the season when hunter-gatherers known today as Bushmen ventured north along this shore in hopes of finding the carcasses of the southern right whales that occasionally wash ashore here.

How the Portuguese fared in these encounters would have been up to them, says Noli. “If they had the wit to trade rather than try to take, there is no reason to believe everybody wouldn’t get along. The few small bands of hunter-gatherers along the river had no population-resource pressures to contend with, and so no reason to be aggressive to the newcomers. On the contrary, a big, strapping Portuguese dom could well have been seen as an attractive prospect for a son-in-law.”

Whatever their final fate, the survivors of the *Bom Jesus* had no inkling of the exquisite irony with which their prayers, uttered so long ago in Lisbon, had been answered. They’d set off on a great journey in search of riches, pledging altars and icons for favor and success. Now here they were, delivered onto a shore of unimaginable wealth—a 185-mile stretch of desert so fantastically rich in high-quality diamonds that in the early 1900s an explorer named Ernst Reuning made a wager with a companion about the amount of time it would take to fill a tin cup with gems found loose in the sand. The job took all of ten minutes.

For long ages the great river had been washing millions, even billions of diamonds down from deposits as far as 1,700 miles inland. Only the hardest, most brilliant, gem-quality stones, some weighing hundreds of carats, survived the journey. They spilled into the Atlantic at the river’s mouth and were washed up the coast, borne by the same cold current that would one day sweep the *Bom Jesus* to its death.

References

Smith, R. (2009, October). Shipwreck in the forbidden zone. National Geographic.

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